In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE(Advanced)-2017

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – V
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	C	B
2.	A	A	B
3.	D	C	B
4.	C	B	A
5.	B	B	A
6.	A	D	C
7.	B	C	D
8.	D	A	A
9.	A	C	D
10.	B	C	B
11.	C	A	C
12.	B	B	A
13.	C	C	C
14.	D	A	D
15.	B	A	B
16.	A	C	A
17.	C	A	C
18.	D	B	D
19.	A	A	A
20.	D	A	B

Physics

PART – I

SECTION – A

1. **A**

Let m_1 and m_2 be the atomic masses of the fragments and v_1 and v_2 their velocities. From the principle of conservation of linear momentum, we have

$$m_1v_1 + m_2v_2 = 0 \qquad \text{or} \qquad \frac{m_1}{m_2} = \left|\frac{v_2}{v_1}\right| = \frac{1}{8}$$

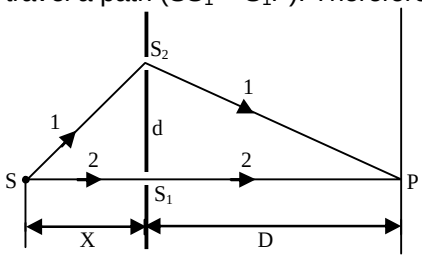
Now $m_1 = A_0(R_1)^{1/3}$ and $m_2 = A_0(R_2)^{1/3}$. Hence

$$\frac{m_1}{m_2} = \left(\frac{R_1}{R_2}\right)^{1/3}$$

or $$\frac{1}{8} = \left(\frac{R_1}{R_2}\right)^{1/3} \quad \text{or} \quad \frac{R_1}{R_2} = \frac{1}{2}$$

2. **A**

Refer to figure. to reach point P, wave 1 has to travel a path ($SS_2 + S_2P$) while wave 2 has to travel a path ($SS_1 + S_1P$). Therefore, when the waves arrive at P, the path difference is



$$\Delta = (SS_2 + S_2P) - (SS_1 + S_1P) \qquad \ldots\ldots\ldots(1)$$

Now, in triangle SS_2S_1, we have

$$SS_2 = (x^2 + d^2)^{1/2} = x\left(1 + \frac{d^2}{x^2}\right)^{1/2}$$

$$= \left(1 + \frac{d^2}{2x^2}\right) \qquad (\because\ d << x)$$

Similarly, $S_2P = (D^2 + d^2)^{1/2} = D\left(1 + \frac{d^2}{2D^2}\right)$ $(\because\ d << D)$

Also $(SS_1 + S_1P) = x + D$. Using these in Eq. (1), we have

$$\Delta = x\left(1 + \frac{d^2}{2x^2}\right) + D\left(1 + \frac{d^2}{2D^2}\right) - (x + D)$$

$$= x + \frac{d^2}{2x} + D + \frac{d^2}{2D} - x - D$$

or $$\Delta = \frac{d^2}{2}\left(\frac{1}{x} + \frac{1}{D}\right)$$

In order to have a dark fringe at P, $\Delta = \frac{\lambda}{2}$. Hence

$$\frac{\lambda}{2} = \frac{d^2}{2}\left(\frac{1}{x} + \frac{1}{D}\right) \qquad \ldots\ldots(1)$$

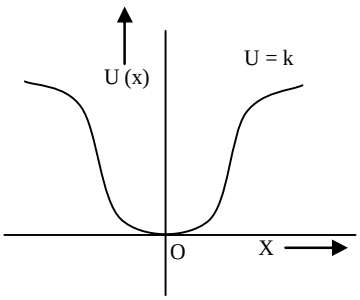
or $$d = \left[\frac{\lambda x D}{x + D}\right]^{1/2} \qquad \ldots\ldots(2)$$

Putting $x = \frac{D}{2}$ in Eq. (2), we find that the correct option is (A).

3. **D**

Figure shows the plot of U(x) versus x. At x = 0, potential energy
$U(0) = k[1 - \exp(0)] = k(1-1) = 0$ and it has a maximum value = k at $x = \pm\infty$ since

$$U(\pm\infty) = k[1 - \exp(-\pm\infty)^2]$$
$$= k(1-0) = k$$



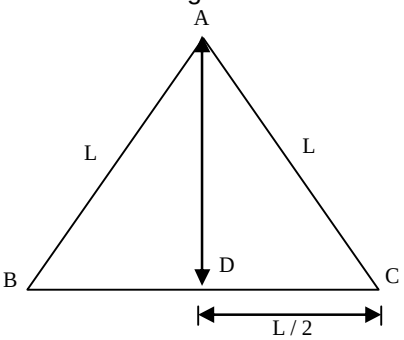
Since the total mechanical energy has a constant value = (k/2), the kinetic energy will be maximum at x = 0 and minimum at $x = \pm\infty$. At x = 0

$$\left(\frac{dU}{dx}\right)_{x=0} = [2\,kx\ \exp(-x^2)]_{\text{at } x=0} = 0$$

Hence the particle is in stable equilibrium at x = 0 (origin) and would oscillate about x = 0 (for small displacements) simple harmonically. Hence (d) is the only correct choice.

4. **C**

Let L be the length of each rod at 0^oC and h_0 be the altitude AD at 0^oC. Then figure.



$$h_0 = \left(L^2 - \frac{L^2}{4}\right)^{1/2} \qquad \ldots\ldots\ldots(1)$$

When the rods are heated to a temperature t, the altitude becomes

$$h = \left[L^2(1+\alpha_1 t)^2 - \frac{L^2}{4}(1+\alpha_2 t)^2\right]^{1/2}$$

Since α_1 and α_2 are much less than unity, we can write $(1+\alpha_1 t)^2 \simeq 1+2\alpha_1 t$ and $(1+\alpha_2 t)^2 \simeq 1+2\alpha_2 t$. Thus

$$h = \left[L^2(1+2\alpha_1 t) - \frac{L^2}{4}(1+2\alpha_2 t\right]^{1/2} \qquad .....(2)$$

Equation (1) and (2), we get

$$\left(L^2 - \frac{L^2}{4}\right)^{1/2} = \left[L^2(1+2\alpha_1 t) - \frac{L^2}{4}(1+2\alpha_2 t\right]^{1/2}$$

Which simplifies to give $\frac{\alpha_1}{\alpha_2} = \frac{1}{4}$, which is choice (C)

5. **B**

$a_{rel} = a\cos 30^o - g\sin 30^o$

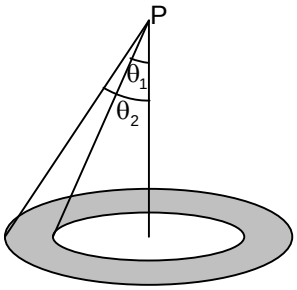
$= 10\sqrt{3} \times \frac{\sqrt{3}}{2} - 5 = 10$

$S = \frac{1}{2}at^2$

$5t^2 = 1$

$t = \frac{1}{\sqrt{5}}$

6. **A**



$v_P = -2\pi G\sigma\left(\cos\theta_1 - \cos\theta_2\right)$

$\theta_1 = 37^o,\ \theta_2 = 45^o$

$\sigma = \frac{M}{\pi R^2(16-9)} = \frac{M}{7\pi R^2}$

$W = -v_P$

7. **B**

$\frac{1}{f_1} = \frac{(\mu_1 - 1)}{R_1} = \frac{0.5}{+14}$

$\frac{1}{f_2} = -\frac{(\mu_2 - 1)}{R_2} = \frac{0.2}{+14}$

$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{0.7}{14} = \frac{1}{20}$

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$u = 40$ cm, $v = 40$ cm

8. **D**

A radioactive nucleus may decay any time after t = 0. The number of neclei at any time t is given by $N = N_0 e^{-\lambda t}$, where $\lambda = \frac{0.693}{T}$

Where T is the half-life. Hence the correct choice is (d)

9. **A**

$\tau = \bar{r} \times \bar{F} = I_{cm}\alpha$

Hence, answer is (i), (ii), i.e. (A).

10. **B**

$W =$ area of loop $= \frac{1}{2}[P_2V_3 - P_2V_2 - P_1V_4 + P_1V_1]$

$= \frac{1}{2}nR[T_3 - T_2 - T_4 + T_1]$

$W = 13.718KJ$

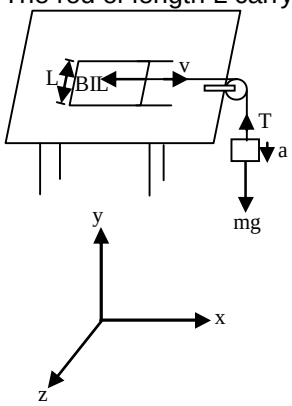
for $1-2, \frac{P_1}{P_2} = \frac{V_1}{V_2}$ & $3 \to 4 \frac{P_2}{P_1} = \frac{V_3}{V_4}$

11. **C**

Refer to figure let v be the velocity of the rod along the positive x-direction at an instant of time and let the magnetic field B act perpendicular to the table along the positive y-direction. The emf induced in the rod is $e = Blv$. Therefore, the induced current is

$$I = \frac{e}{R} = \frac{BLv}{R} \qquad \text{.....(1)}$$

The rod of length L carrying a current I in magnetic field will experience a force



Along the negative x-direction. Since the rod is massless, this force will also be equal to the tension T in the string acting along the positive x-direction i.e. T = f = BIL.

Let α be the acceleration of mass m moving in the downward direction, then

Ma = net force acting on m = mg – T = mg – F

or $a = g - \frac{F}{m}$

Using (1) and (2) in (3), we have

$$a = g - \frac{BIL}{m} = g - \frac{B \times BL^2 v}{mR}$$

$$= g - \frac{B^2 L^2 v}{mR}$$

So the correct choice is (c)

12. **B**

When the velocity of the rod is half the terminal velocity, then $a = \frac{g}{2}$.

13. **C**

From Snell's law and n is negative.

14. **D**

$|n_1|\lambda_1 = |n_2|\lambda_2$

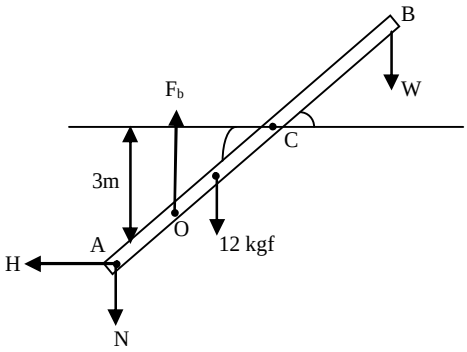
15. **B**

Mass per unit length of rod is 2 kg.

Mass of submerged portion of rod is 10 kg and its volume is $\frac{10}{500}m^2$.

Buoyant force $= F_B = \frac{10}{500} \times 1000 = 20\text{kgf}$

16. **A**



Let N and H be the vertical and horizontal reactions of the hinge on the rod. Considering horizontal and vertical translation equilibrium of rod.

N + 12 + W = 20, where W is the weight to be attached at end

H = 0

Considering the rotation equilibrium of rod about A

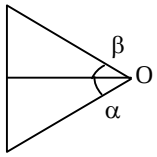
$20 \times 2.5\cos\theta + 12 \times g \times 3\cos\theta + W \times 6\cos\theta = 0$

$6w = 50g - 36g = 14g$

$W = \frac{7}{3}g = \frac{7}{3}\text{kgf}$

$N = 8 - \frac{7}{3} = \frac{17}{3}\text{kgf}$

17. **C**



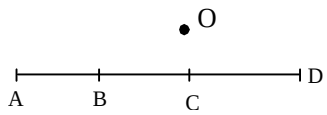
Potential at O,

$$V = K\lambda \ln\left(\frac{\sec\alpha + \tan\alpha}{\sec\beta - \tan\beta}\right)$$

In case P, $\alpha = \beta = 60$

$$V = 3 \times V_1 = 3K\lambda \ln\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right) = 6K\lambda \text{Ln}\left(2+\sqrt{3}\right)$$

In case Q,



$$V_{BC} = 2K\lambda \ln\left(2+\sqrt{3}\right)$$

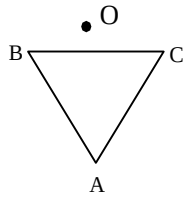
$$V_{AB} = V_{CD} = K\lambda \ln\left(\frac{\sqrt{28}+3\sqrt{3}}{2+\sqrt{3}}\right)$$

$$V = V_{BC} + V_{AB} + V_{CD} = 2K\lambda \ln\left(2\sqrt{7}+3\sqrt{3}\right)$$

Case – R,

$$V = \frac{Kq}{r} = \frac{K\lambda 2\pi r}{r} = K\lambda \cdot 2\pi$$

Case S,



$$V_{AB} = V_{AC} = K\lambda \ln\left(2+\sqrt{3}\right)$$

$$V = 4K\lambda \ln\left(2+\sqrt{3}\right)$$

18. **D**

From the concept of floating level will remain same if object floats and will down if it sinks.

19. **A**

Area = work, $\Delta U = nC_v\Delta T$ and $\Delta Q = \Delta U + W = \int nCdT$

(P) Work is negative and $\Delta U > 0$

(Q) For Q, $\Delta U < 0$, $\therefore \Delta T < 0$, $\Delta V < 0$ from $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

So, answer is (2, 5).

(R) $\Delta V > 0$, $W > 0$, but $\Delta T < 0$, $\therefore \Delta U < 0$

Answer is 3.

(S) $W < 0$, $\Delta U < 0$

Answer is (2, 5).

20. **D**

Use the property of images formed by mirror and lenses.

Chemistry

PART – II

SECTION – A

1. **C**

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

11 12

$11-x \quad 12-3x \quad 2x$

at equilibrium $12-3x=6$

$x=2$

$n_{H_2}=6 \quad n_{N_2}=9 \quad n_{NH_3}=4$

$PV=\left(n_{N_2}+n_{H_2}\right)RT$

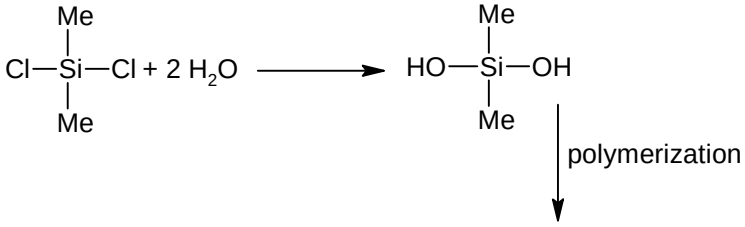
$P\times(20-3.58)=15\times0.0821\times300$

$P=22.5$ atm

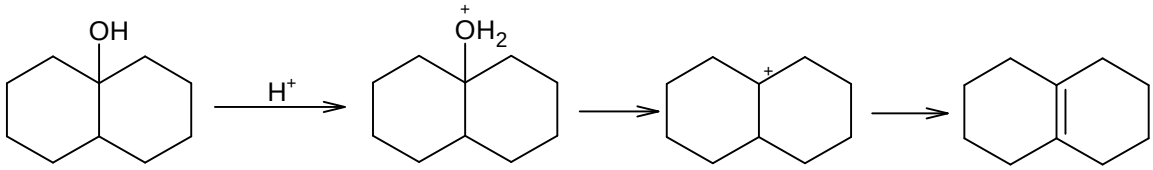
2. **A**

Exothermic reaction mean products are having lesser energy than reactants and $k_1<k_2 \Rightarrow$ activation energy corresponding k_1 is more than k_2.

3. **C**



4. **B**



5. **B**

Anode $Ag(s) \rightarrow Ag^+(aq)+e^-$

Cathode $Ag^+(aq)+e^- \rightarrow Ag(s)$

$$E_{cell}=E^0_{cell}-\frac{0.0591}{n}\log\frac{\left[Ag^+\right](anode)}{\left[Ag^+\right]cathode}$$

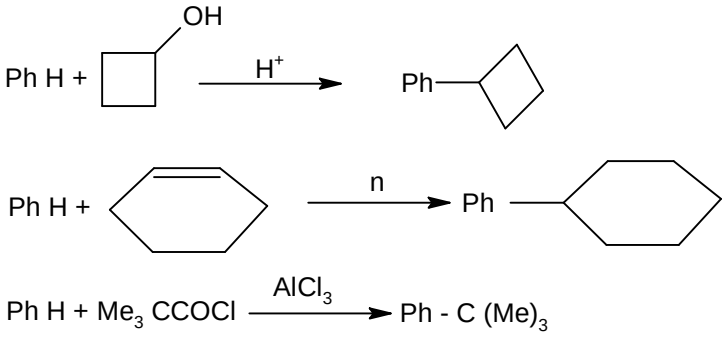
At equilibrium, $E_{cell}=0$

K_{sp} of AgBr = $\left[Ag^+\right]\left[Br^-\right]$

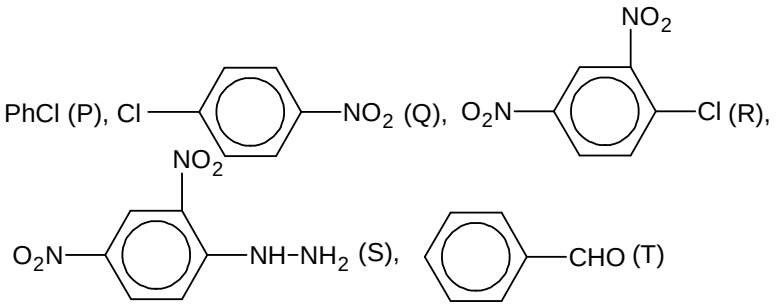
6. **D**

The bond order in O_2 is 2 while in $H_2S_2O_8$, H_2O_2 have one peroxide bond each while in $O_2[BF_4]$ the bond order is 2.5.

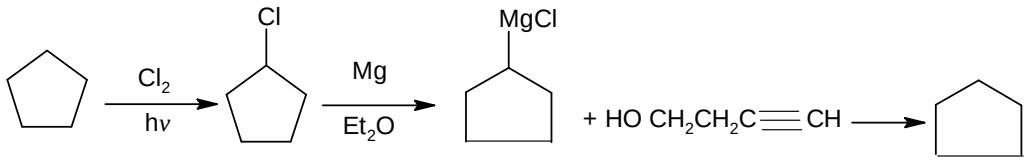
7. **C**



8. **A**



9. **C**



1 mole 2 mole $n = \frac{3.5}{70} = 0.05$

0.5 moles of compound $HOCH_2CH_2–C\equiv CH$ give = 1 mole 

0.05 moles of compound give = 0.1 mole  =7 gm

10. **C**
Fact.

11. **A**

$$2NaHCO_{3(s)} \rightleftharpoons Na_2CO_{3(s)} + \underset{P}{CO_{2(g)}} + \underset{P}{H_2O_{(s)}}$$

$K_p = P^2 = 0.23 \Rightarrow P = 0.48 \Rightarrow P_{total} = 2P = 0.96 atm$

12. **B**

$$2NaHCO_3 \rightleftharpoons Na_2CO_{3(s)} + \underset{2.1-P_1}{CO_{2(g)}} + \underset{0.94-P_1}{H_2O_{(g)}}$$

$(2.1-P_1)(0.94-P_1) = 0.23$

$P_1 = 0.77$ atm P_{CO_2} 1.33 atm $P_{H_2O} = 0.19$ atm

13. **C**

$\Delta T_b = iK_b.m$

$6 = i \times 2 \times 1$

$i = 3$

Complex will dissociate in three fragments only so complex is $\left[CO(NH_3)_5\,Cl\right]Cl_2$

14. **A**

Moles of AgCl = 1

Moles of AgCl / mole of complex

= ionizable Cl^- in complex = 1 $\left[CO(NH_3)_4\,Cl_2\right]Cl$

15. **A**

$(NH_4)_2C_2O_4 \longrightarrow (CN)_2 + 4H_2O$

White crystalline solid Cyanogen gas (Pseudohalogen gas)

16. **C**

Fact.

17. **A**

(P) $\Delta E = nC_V\Delta T$

$\Delta H = nC_P\Delta T$ and $\Delta T = 0$

(Q) $Q_{rev} = 0$

(R) Δ State functions are zero.

(S) $\Delta V = 0$

18. **B**

(P) $\sigma 1s^2,\ \sigma^* 1s^2,\ \sigma 2s^2,\ \sigma^* 2s^2,\ \sigma 2p_x^2,\ \pi 2p_y^2,\ \pi 2p_z^2,\ \pi^* 2p_y^1,\ \pi^* 2p_z^1$

(Q) $\sigma 1s^2,\ \sigma^* 1s^2,\ \sigma 2s^2,\ \sigma^* 2s^2,\ \pi 2p_y^1,\ \pi 2p_z^1$

(R) $\sigma 1s^2,\ \sigma^* 1s^2,\ \sigma 2s^2,\ \sigma^* 2s^2,\ \pi 2p_y^2,\ \pi 2p_z^2$

(S) $\sigma 1s^2,\ \sigma^* 1s^2,\ \sigma 2s^2,\ \sigma^* 2s^2,\ \pi 2p_y^2,\ \pi 2p_z^2,\ \sigma 2p_x^2$

19. **A**

(P) CH_3COONa, NH_3, $NaHCO_3$, HCOONa will react with HCl and form a weak acid, so can make buffer.

(Q) NH_4Cl and $NaHCO_3$ will react with NaOH to form buffer.

(R) Will form an ammonia complex.

(S) All weak acids and weak base can form buffer solution.

20. **A**

(P) $\Delta n = 0$

(Q) $\Delta n = -2$

(R) $\Delta n = +1$

(S) $\Delta n = +2$

Mathematics

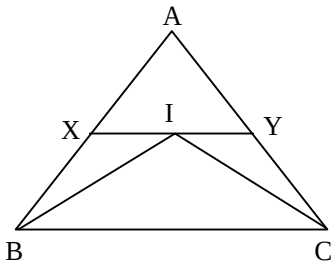
PART – III

SECTION – A

1. **B**

From the relation $(BI)^2 = BX.BA$, we see that BI is a tangent to the circle passing through A, X, I at I.

Hence $\angle BIX = \angle BAI = A/2$ .....(i)



Similarly $CI^2 = CY.CA$ we have

$\angle CIY = \angle CAI = A/2$ .....(ii)

It is known that

$\angle BIC = 90^o + A/2$ .....(iii)

From (i), (ii) & (iii) we obtain

$\frac{A}{2} + \frac{A}{2} + \left(90^o + \frac{A}{2}\right) = 180^o$

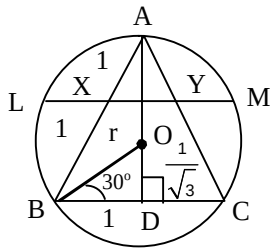
$\Rightarrow A = 60^o$

2. **B**

Let the circumcircle of the equilateral ΔABC have centre O(0, 0) & radius r. Join A to D, the midpoint of BC, then AD passes through O & is perpendicular to BC. Draw OB. Let LM cut the sides AB & AC of ΔABC at X & Y. Then $\angle XYM$ is parallel to BC

$OD = \tan 30^o = 1/\sqrt{3}$, $D = \left(0, -1/\sqrt{3}\right)$, $A\left(0, \frac{2}{\sqrt{3}}\right)$ & $B\left(-1, -1/\sqrt{3}\right)$

Equation of circle is $x^2 + y^2 = \frac{4}{3}$



X is midpoint of AB $\left(\frac{-1}{2}, \frac{-10}{2\sqrt{3}}\right)$

on solving $x^2 + y^2 = \frac{4}{3}$ with line $y = -\frac{1}{2\sqrt{3}}$ we get

$x^2 + \left(\frac{1}{2\sqrt{3}}\right)^2 = \frac{4}{3}$

$\therefore$ x coordinate of L & M is $\frac{\sqrt{5}}{2}$ & $\frac{-\sqrt{5}}{2}$ respectively

$LM = \sqrt{5}$

3. **B**

$\log_5 50 - \log_5 2 = 2$

4. **A**

Using $\cos(\theta + \phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$ in each term & simplifying we have

$y(x) = \lambda \sin(x + \alpha)$ for some constant λ & α. So zeroes of y(x) are of the form $x + \alpha = n\pi$, $n \in I$.

$y(x_1) = y(x_2) \Rightarrow x_1 - x_2 = n\pi$.

5. **A**

$a_n + \sqrt{2}b_n = \left(2+\sqrt{2}\right)^n$

$a_n - \sqrt{2}b_n = \left(2-\sqrt{2}\right)^n$

$a_n = \frac{1}{2}\left[\left(2+\sqrt{2}\right)^n + \left(2-\sqrt{2}\right)^n\right]$

$b_n = \frac{\left(2+\sqrt{2}\right)^n - \left(2-\sqrt{2}\right)^n}{2\sqrt{2}}$

$\lim_{n\to\infty}\frac{a_n}{b_n} = \lim_{n\to\infty}\sqrt{2}\left(\frac{1+\left(\frac{2-\sqrt{2}}{2+\sqrt{2}}\right)^n}{1-\left(\frac{2-\sqrt{2}}{2+\sqrt{2}}\right)^n}\right)$

$= \sqrt{2}\left(\frac{2-\sqrt{2}}{2+\sqrt{2}} < 1\right)$

6. **C**

Using De Moivre's theorem

$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

$\Rightarrow \sin n\theta = \sum_{k=0}^{\left[\frac{n-1}{2}\right]}(-1)^k\ {}^nC_{2k+1}\cos^{n-2k-1}\theta\sin^{(2k+1)}\theta$

$\sin(2n+1)\theta = \sum_{k=0}^{n}(-1)^k\ {}^{2n+1}C_{2k+1}\cos^{2n-2k}\theta\sin^{2k+1}\theta$

$= \tan\theta\cos^{2n+1}\theta\sum_{k=0}^{n}(-1)^k.{}^{2n+1}C_{2k+1}(\tan\theta)^{2k}$

$\sum_{k=0}^{n}(-1)^k\ {}^{2n+1}C_{2k+1}\tan^{2k}\theta = 0$

for $\theta = \frac{j\pi}{2n+1}$, $1 \le j \le n$

so $\tan^2\frac{j\pi}{2n+1}, 1 \le j \le n$ are the roots of $\sum_{k=0}^{n}(-1)^k.{}^{2n+1}C_{2k+1}.x^k = 0$

$\because a_n$ & b_n are the sum and the product of the roots respectively, we have

$a_n = {}^{2n+1}C_2 = n(2n+1)$

$b_n = {}^{2n+1}C_{2n} = 2n+1$

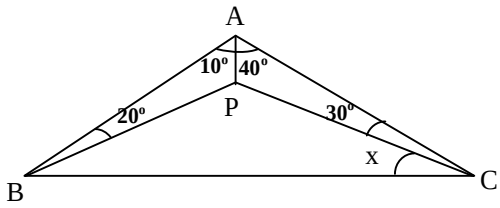
$\therefore \frac{a_n}{b_n} = n$

7. **D**

${}^{100}C_1\sin\theta + {}^{100}C_2\sin 2\theta + ... + {}^{100}C_{100}\sin 100\theta$ = Imaginary part of $\left(1+e^{i\theta}\right)^{100}$

= Imaginary part of $(1+\cos\theta+i\sin\theta)^{100} = 2^{100}\left(\cos\frac{\theta}{2}\right)^{100}\sin 50\theta$

8. **A**



Let $x = \angle PCB$ then $\angle PBC = 80^\circ - x$

By sin law

$$1=\frac{PA}{PB}.\frac{PB}{PC}.\frac{PC}{PA}$$

$$=\frac{\sin\angle PBA}{\sin\angle PAB}.\frac{\sin\angle PCB}{\sin\angle PCB}.\frac{\sin\angle PAC}{\sin\angle PCA}$$

$$=\frac{\sin 20^\circ \sin x^\circ \sin 40^\circ}{\sin 10^\circ \sin\left(80^\circ - x\right)\sin 30^\circ}$$

$$=\frac{4\sin x \sin 40^\circ \cos 10^\circ}{\sin\left(80^\circ - x\right)}$$

$$1=\frac{2\sin x\left(\sin 30^\circ + \sin 50^\circ\right)}{\sin\left(80^\circ - x\right)}$$

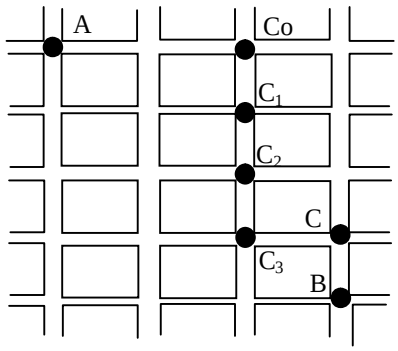
$$=\frac{\sin x\left(1+2\cos 40^\circ\right)}{\sin\left(80^\circ - x\right)}$$

& so $2\sin x \cos 40^\circ = \sin\left(80^\circ - x\right) - \sin x = 2\sin\left(40^\circ - x\right)\cos 40^\circ$

$\Rightarrow x = 40^\circ - x$ or $x = 20^\circ$

$\therefore \angle ACB = 50^\circ$.

9. **D**



The probability that the student passes through C is the sum from $i = 0$ to 3 of the probabilities that he enters intersection C_i in the adjoining figure & goes east. The number of paths from A to C_i is ${}^{2+i}C_2$ because each such path has 2 eastward block segments & they can occur in any order. The probability of taking any one of these paths to C_i and then going east is $\left(\frac{1}{2}\right)^{3+i}$ because there are $3+i$ intersection along the way (including A & C_i) where an independent choice with probability ½ is, made

$$\therefore \sum_{i=0}^{3} {}^{2+i}C_2\left(\frac{1}{2}\right)^{3+i} = \frac{21}{32}$$

10. **B**

$$\lim_{x\to 0}\left\{\frac{f(x)-f(0)}{x-0}+\frac{f\left(\frac{x}{2}\right)-f(0)}{x-0}+\frac{f\left(\frac{x}{3}\right)-f(0)}{x-0}+.....+\frac{f\left(\frac{x}{k}\right)-f(0)}{x-0}\right\}.$$

$$=\left(1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{k}\right)f'(0)$$

11. **C**

$$\frac{f'(x)}{f(x)} = \frac{f'(-x)}{f(-x)}$$

Integrating we get

$f(x).f(-x) = c$

$f(0) = 3 \Rightarrow c = 9$

$f(x)f(-x) = 9$

12. **A**

$$I = \int_{-51}^{51} \frac{dx}{3 + f(x)}$$

Let $x = -t$

$$I = \int_{-51}^{51} \frac{dt}{3 + f(-t)}$$

$$2I = \int_{-51}^{51} \frac{6 + f(x) + f(-x)}{18 + 3\left(f(x) + f(-x)\right)}$$

$$\Rightarrow 2I = \int_{-51}^{51} \frac{dx}{3}$$

$I = 17$

13. **C**

14. **D**

(Soln: 13 – 14)

$$\frac{s-a}{s} = \frac{1}{2} \Rightarrow b + c = 3a \Rightarrow CA + BA = 12$$

A lies on ellipse whose foci are B & C centre of ellipse is (4, 4)

Length of major axis = 12

$$12e = 4 \Rightarrow e = \frac{1}{3}$$

Length of minor axis $= 8\sqrt{2}$.

(i) PQ is the latus rectum of ellipse

$$\therefore \text{Area} = 2\left(\frac{1}{2} \times K_1 e \times \frac{k_2^2}{k_1}\right) = ek_2^2 = \frac{32}{3} \text{sq.units}$$

(ii) ΔPBC is et $L^d at^c$

(iii) Equation of Hyperbola is

$$\frac{(x-4)^2}{4} - \frac{(y-4)^2}{4\left(e^2 - 1\right)} = 1$$

It passes through $(0,2) \Rightarrow e = \frac{2}{\sqrt{3}}$

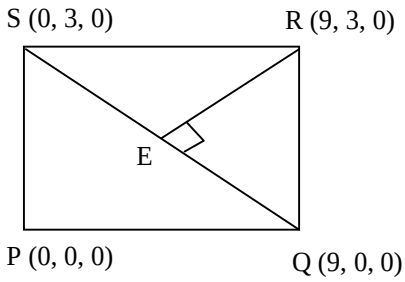
15. **B**

Equation of lines QS in 2D will be $x+3y-9=0$

$RE=\frac{9}{\sqrt{10}}$ & $E\left(\frac{81}{10},\frac{3}{10},0\right)$

So point T will be

$\therefore PT=\frac{3}{5}\sqrt{205}$



16. **A**

Shortest distance between the lines

$\bar{r}=\bar{a}+\lambda\bar{\alpha}$ & $\bar{r}=\bar{b}+\mu\bar{\beta}$

Is given by $\frac{\left|\left(\bar{a}-\bar{b}\right).\left(\bar{\alpha}\times\bar{\beta}\right)\right|}{\left|\bar{\alpha}\times\bar{\beta}\right|}=3\sqrt{\frac{10}{19}}$

17. **C**

B is cofactor of A

$\therefore |B|=|A|^2$

$\det(AB)=|A|^3$

$|A|=3abc-a^3-b^3-c^3$

$=18$

$|AB|=(18)^3$

$\alpha+\beta+\gamma=tr(|B|)=ab+bc+ca-a^2-b^2-c^2$

$=3(ab+bc+ca)-(a+b+c)^2$

$=3\times5-9=6$

18. **D**

Let the line be $\frac{x}{\cos\theta}=\frac{y}{\sin\theta}=r$

$r_1\cos\theta=\alpha$

$r_2\sin\theta=\beta$

$r_3(\sin\theta+\cos\theta)=8$

$r_1r_2r_3=48\sqrt{2}$

$f(\alpha,\beta)\le0$

$\Rightarrow\left|\frac{\beta}{\alpha}-\frac{3}{2}\right|+(3\alpha-2\beta)+\sqrt{e^{\alpha}+2\beta-2e-6}\le0$

$\Rightarrow\left|\frac{\beta}{\alpha}-\frac{3}{2}\right|+(3\alpha-2\beta)^6+\sqrt{e^{\alpha}+2\beta-e^2-6}=0$

$\frac{\beta}{\alpha}-\frac{3}{2}=0,\ 3\alpha-2\beta=0$ & $e^{\alpha}+2\beta=e^2+6$ .....(1)

$\beta = \frac{3\alpha}{2}, \beta = \frac{3\alpha}{2},$

Putting $\beta = \frac{3\alpha}{2}$ in (1)

$e^{\alpha} + 3\alpha = e^2 + 6$

$\Rightarrow \alpha = 2, \beta = 3$

$\Rightarrow \quad r_1 = \frac{2}{\cos\theta}$

$r_2 = \frac{3}{\sin\theta}$

$r_3 = \frac{8}{\sin\theta + \cos\theta}$

$\frac{48}{\cos\theta\sin\theta(\sin\theta + \cos\theta)} = 48\sqrt{2}$

$\cos\theta\sin\theta(\sin\theta + \cos\theta) = \frac{1}{\sqrt{2}}$

$\Rightarrow \quad \theta = \frac{\pi}{4}$

$r_1 = 2\sqrt{2},\ r_2 = 3\sqrt{2},\ r_3 = 4\sqrt{2}$

19. **A**

(P) $\lim\limits_{y\to\infty} \frac{y - \tan 1}{y - 1} = 1$

(Q) $4ab = (a+b)^2 - (a-b)^2$

$4ab = (2 + 2\sin^2 2\theta)^2 - 4^2 \sin^2 2\theta$

$ab = (1 + \sin^2 2\theta)^2 - 4\sin^2 2\theta$

$\Rightarrow ab = (1 - \sin^2 2\theta)^2$

$(ab)_{max} = 1$

(R) $\cos\alpha = \frac{\frac{1}{2}}{2} = \frac{1}{4}$ (from Δ OED)

$\theta = \pi - 2\alpha$

$BD = 2\cot\frac{\theta}{2} = 2\cot\left(\frac{\pi}{2} - \alpha\right)$

$= 2\tan\alpha$

$= 2\sqrt{15}$

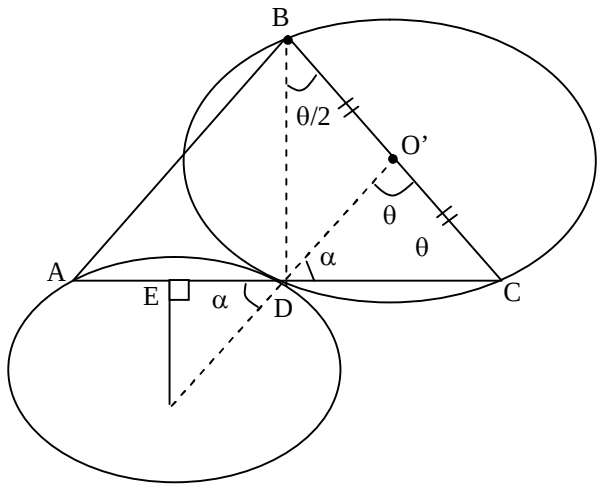
Area of $\Delta ABC = \frac{1}{2}(AC)(BD)$

$= \frac{1}{2} \times 3 \times 2\sqrt{15}$

$= 3\sqrt{15}$

$= \sqrt{135}$

$\frac{N^2}{15} = 9$



(S) $\lim\limits_{n\to\infty}\dfrac{n^{2/3}\sin(n!)}{n\left(1+\dfrac{1}{n}\right)}=\lim\limits_{n\to\infty}\dfrac{1}{n^{1/3}}.\dfrac{\sin(n!)}{1+\dfrac{1}{n}}$

$=\dfrac{1}{\infty}\left(\dfrac{\text{a number between } -1\&1}{1}\right)$

$=0$

20. **B**

(P)

$I_1=\int_1^e x\left(\dfrac{1+x}{x}\right)(x+\ln x)^{100}\,dx$

$=\dfrac{e(e+1)^{101}-1}{101}-\dfrac{1}{101}\int_1^e(x+\ln x)^{101}dx$

$I_2=\dfrac{1}{e}\int_1^e(t+\ln t)^{101}dt \qquad \text{let} \quad e\sin x=t$

$\Rightarrow I_1+\dfrac{eI_2}{101}=\dfrac{e(e+1)^{101}-1}{101}$

$\Rightarrow K=1$

(Q)

$KI(k)=\int_0^\infty\dfrac{k\ln x}{x^2+kx+k^2}dx$

$x=kt$

$kI(k)=\int_0^\infty\dfrac{\ln Kt}{t^2+t+1}dt$

$kI(k)-I(1)=\ln k\int_0^\infty\dfrac{dt}{1+t+t^2}$

$=\ln k\int_0^\infty\dfrac{dt}{\left(t+\dfrac{1}{2}\right)^2+\left(\dfrac{\sqrt{3}}{2}\right)^2}$

$\Rightarrow\dfrac{\pi}{\sqrt{3}}=(\ln k)\dfrac{2}{\sqrt{3}}.\dfrac{2\pi}{6}$

$k=e^{3/2}$

(R)

$9f(3) \le \int_0^3 x^2f'(x)+\int_0^3(f(x))^2\,dx+k$

$\le\left[x^2b(x)\right]_0^3-\int_0^3 2xf(x)dx+\int_0^3(f(x))^2dx+k$

$\Rightarrow \int_0^3\left(2xf(x)-(f(x))^2\right)dx\le k$

$2xf(x)-f^2(x)=x^2-(f(x)-x)^2\le x^2$

$\int_0^3 2xf(x)-f^2(x)dx\le\int_0^3 x^2dx$

$\int_0^3\left(2xf(x)-(f(x))^2\right)dx\le 9$

$K_{min}=9$

(S)

$$\log_a x = t$$

$$\Rightarrow \quad a^t = x$$

$$\Rightarrow \quad dx = a^t \ln a\, dt$$

$$\therefore \quad \int_1^a x\, a^{-[\log_a x]} dx = \ln a \int_0^1 a^t . a^{-[t]} . a^t dt$$

$$= \quad \ln a \int_0^1 a^{2t} dt = \frac{a^2 - 1}{2} = \frac{e - 1}{2}$$

$$\Rightarrow \quad a = \sqrt{e}$$